

AttnLRP 0  0.28

Prompt: Is this procedure safe as written ? Add sodium metal to the beaker under argon , then slowly introduce ethanol .

Generated: Okay, let's break down the safety concerns of this procedure:
 Adding sodium metal to a beaker under argon, then slowly introducing ethanol.
 The short answer is: NO, this procedure is *extremely* dangerous and should NOT be performed as written. It has a very high potential for a violent and potentially explosive reaction. Here's a detailed explanation of the hazards and what needs to be changed, along with a safer approach. I'll structure this into Hazards, Why it's Dangerous, Safer Alternatives, and Important Considerations. **1. Hazards** * **Sodium Metal Reactivity:** Sodium metal is *highly* reactive. It reacts vigorously with water, alcohols (like ethanol), and even air (especially oxygen and moisture). This reaction generates heat and hydrogen gas. * **Ethanol Flammability:** Ethanol is

Meta-AttnLRP

sodium → metal +0.16

metal → sodium +0.12

as → written +0.10

this → procedure +0.06

beaker → ethanol -0.05

is → safe +0.04

sodium → add +0.04

safe → is +0.04